

Arpit Kumar Rai

3rd Year Undergraduate

Department of Computer Science and Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2020 - Present	B.Tech	Indian Institute of Technology Kanpur	9.9/10
2020	CBSE(XII)	Sunbeam English School, Varanasi	96.4%
2018	ICSE(X)	Children College, Azamgarh	97.6%

Scholastic Achievements

- Awarded **Academic Excellence Award** for two consecutive years **2020** and **2021** for exceptional academic performance
- Secured **All India Rank 155** and **Rank 5** in **IIT Kanpur Zone** in **JEE Advanced 2020** among 1.2 Lakh candidates
- Secured **All India Rank 603** in **Joint Entrance Exam (JEE) Mains 2020** among more than 1 million candidates
- Secured **All India Rank 562** and awarded fellowship in **Kishore Vaigyanik Protsahan Yojana (KVPY) 2019**
- Among **National top 1%** in National Standard Examination in Physics (**NSEP**) and qualified for **INPhO** and **INAO**
- Highly active participant on **Codeforces** contests, with the highest rating achieved of **1779**, handle - Arpitkr

Publications

Arpit Kumar Rai, Arijit Khan, Naheed A. Arafat, B. Ghosh. “Neighbourhood-Based Hypergraph Core Decomposition”
Under review **ACM SIGMOD - 2023**

Work Experience

- **Nanyang Technical University (NTU), Singapore** (Mar’22-July’22)
Research Internship: Prof. Arijit Khan, School of Computer Science and Engineering, NTU
 - **Objective:** To develop a novel approach for **hypergraph core decomposition** using neighborhood-based parameters
 - Implemented **Peel**, **E-Peel** and **Local-core** algorithms to compute **neighborhood-based cores in a hypergraph**
 - Came up with **Local-core** specific optimizations achieving a **maximum speedup** of over **500%** on **congress** dataset
 - Performed **experiments** demonstrating the use of neighbourhood-based hypergraph decomposition in **intervening infection/ misinformation** spread to be more **natural** and **accurate** as opposed to **degree-based decomposition**

Key Projects

- **IITK Cycling App** | 🌐 Arpitkr/IITK-Cycling (Jan’22-Apr’22)
Course Project. Supervisor: Dr. Indranil Saha, Department of Computer Science and Engineering, IIT Kanpur
 - Successfully developed and deployed a **fullstack web application** to facilitate **cycle-renting** in IIT Kanpur campus
 - Collaborated in a team of ten members to deliberate on the features and implementation details of the web application
 - Employed a **real-time database** using **MongoDB Atlas**, with **NodeJS** for backend and **ReactJS** on the front-end
 - Learnt relevant development practices such as **Requirement Documentation**, **Design Documentation** and **Testing**
- **SAT Solver** | 🌐 Arpitkr/SAT-solver (Jan’22- Feb’22)
Course Project. Supervisor: Dr. Subhajit Roy, Department of Computer Science and Engineering, IIT Kanpur
 - Implemented a **SAT Solver** in C++ language using the **Davis–Putnam–Logemann–Loveland (DPLL)** algorithm
 - Employed **advanced optimizations** such as **unit propagation** to eliminate **redundant variable assignments**
 - Correctly computed **satisfiable models** for **propositional logic formula** with over **150 variables** and **650 clauses**
- **IITK Coin** | 🌐 Arpitkr/IITK-Coin (May’21- July’21)
Programming Club, Science and Technology Council, IIT Kanpur
 - Developed a reward-based **centralised pseudo-currency** system in **GoLang** built exclusively for IITK community
 - Implemented security intensive features, password authentication by **hashing & salting** and session validation by **JWTs**
 - Incorporated measures to maintain **ACID** properties of database, prevent deadlock and ensure consistent transactions

Technical Skills

Proficient	Familiar	Frameworks/Libraries	Tools/Utilities
C, C++, Python, Javascript	Golang, Julia, Verilog, SQL	NodeJS, ExpressJS, ReactJS, NumPy, Pandas	Git, L ^A T _E X, Bash

Relevant Courses

Data Structure & Algorithms	Fundamentals of Computing	Discrete Mathematics
Probability & Statistics	Software Development and Operations	Computer Architecture
Real Analysis & Advanced Calculus	Linear algebra & Differential Equations	Introduction to Electronics
Operating Systems	Advanced Algorithm	Quantum Computing

Volunteer Work

- Mentored over **60 students** in the project **Unmasking CP** to start their journey in **Competitive Programming (2022)**
- Introduced the students with **Julia** and **Golang** under the project **6 Weeks 6 Languages** by **Programming Club (2022)**

Miscellaneous

- **Won** the **September** edition of **Sports Programmer of The Month** organised by Programming Club, IIT Kanpur
- Implemented a **k-sudoku pair solver** and **maximal sudoku pair generator** using **pySAT** module of Python